

Exercise Sheet 3

The Gradient(descent) Method

The gradient method is used to minimize a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ according to the iteration rule

$$x^{k+1} = x^k - \alpha \nabla f(x^k).$$

First, we consider the naive implementation with constant step size α .

Task 1 Implementation of the gradient method

Implement the gradient method according to the following algorithm:

Algorithm 1 GradientDescent($\nabla f, x^0, \text{tol} = 10^{-6}, \text{maxit} = 1000$)

```
1:  $k \leftarrow 0$ 
2:  $\text{res} \leftarrow 1$ 
3:  $\text{res\_list} \leftarrow []$ 
4: while  $\text{res} > \text{tol}$  and  $k < \text{maxit}$  do
5:    $x^{k+1} \leftarrow x^k - \alpha \nabla f(x^k)$ 
6:    $\text{res} \leftarrow \|\nabla f(x^{k+1})\|$ 
7:    $\text{res\_list}[k] \leftarrow \text{res}$ 
8:    $k \leftarrow k + 1$ 
9: end while
10: return  $x^{k+1}, \text{res\_list}, \text{x\_list}$ 
```

The function should return the final result x^{k+1} , a vector res_list with all residuals and a list x_list of all intermediate results x^k .

Tip: Implement a class `GradientDescent(Solver)` that inherits from an abstract solver class `Solver` so that you can easily implement additional solvers in the future.

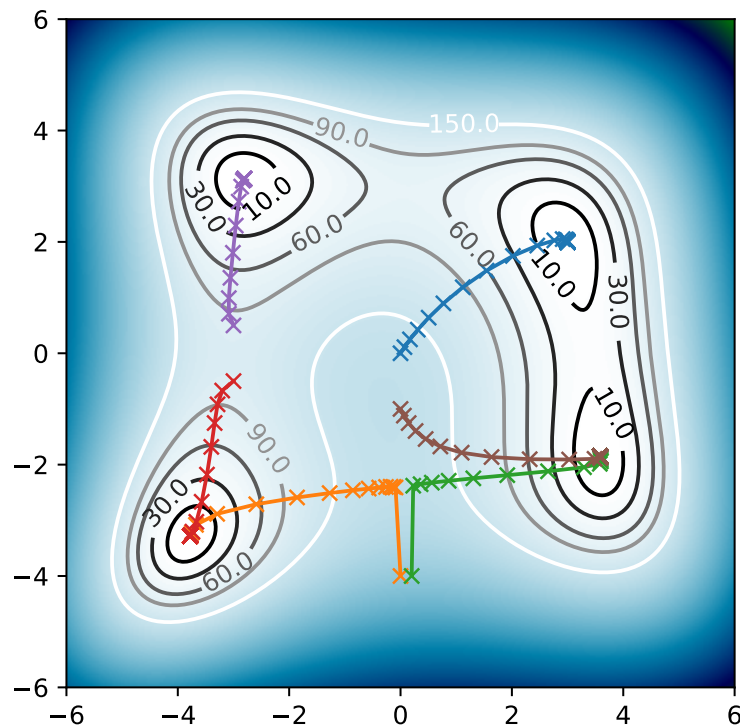


Fig. 1: A contour plot of Himmelblau function and paths from various values for x_0 to the respective solution.

task 2. application to the Himmelblau function

Apply your method from task 1 to the Himmelblau function

$$f_1(x_1, x_2) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$$

and plot the solution steps during the gradient descent (see Fig. 1).

- Use $\alpha = 0.01$ and compare the solution for the starting values $x^0 = (0, 0)$, $x^0 = (2, -3)$, $x^0 = (-3, -1)$ and $x^0 = (-3, 1)$. What do you observe?
- Compare different step sizes α . If your program diverges, reduce the step size.
- Implement a decreasing step size

$$\alpha_k = \frac{\alpha}{\sqrt{k+1}}$$

What do you observe?

- Plot the residual over the iteration number with logarithmized y -axis.